

Lesson 4

Managing Physical Assets

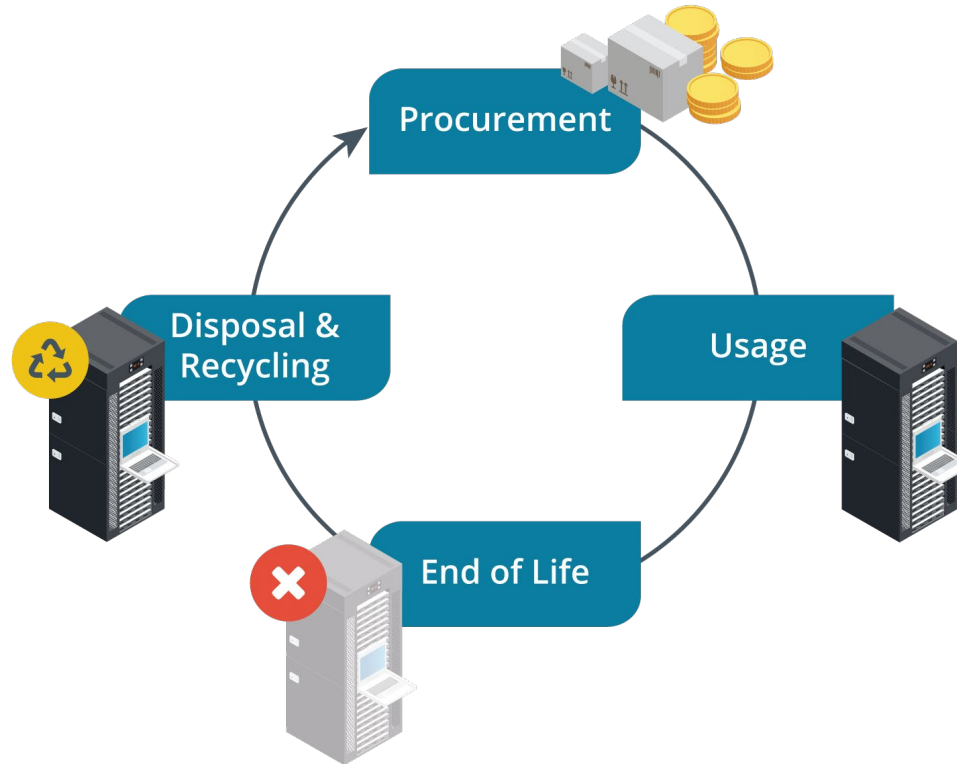
Topic 4A

Understand Asset Management Concepts

Exam Objectives Covered

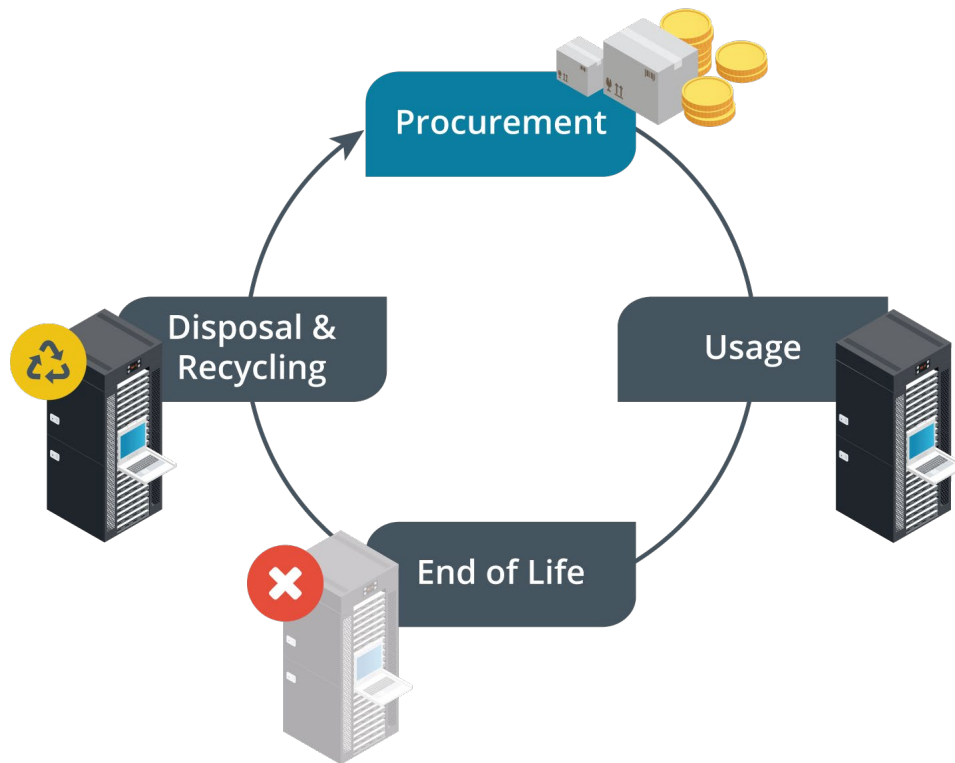
- 2.7 Explain the importance of asset management and documentation

Server Lifecycle



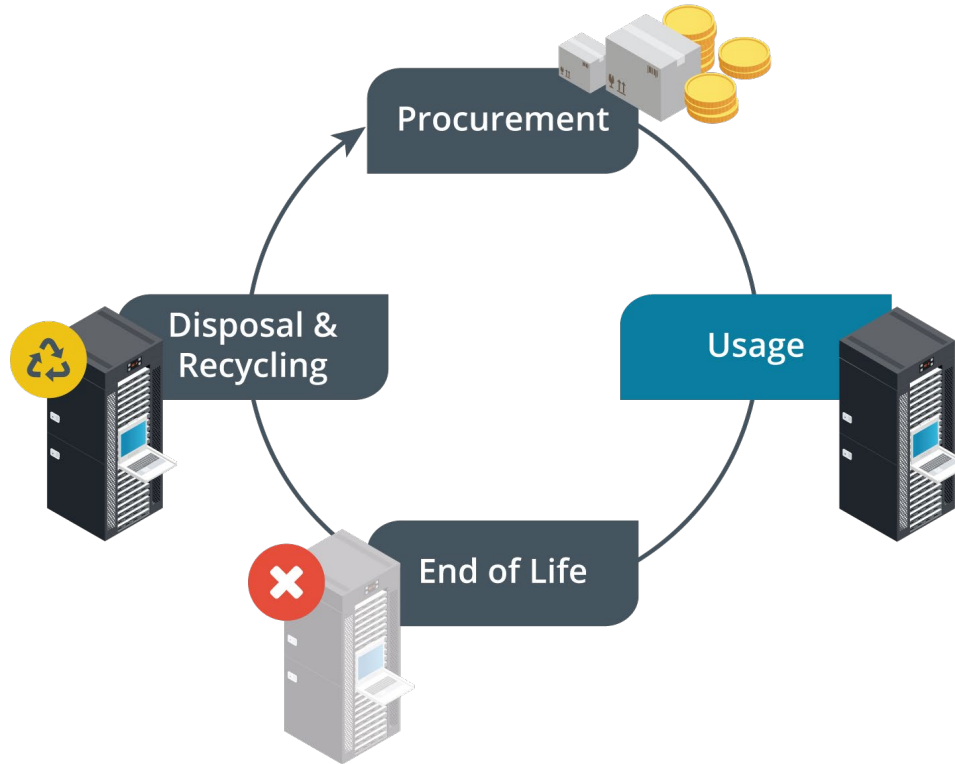
Server lifecycle illustration. (Images © 123RF.com)

Procurement



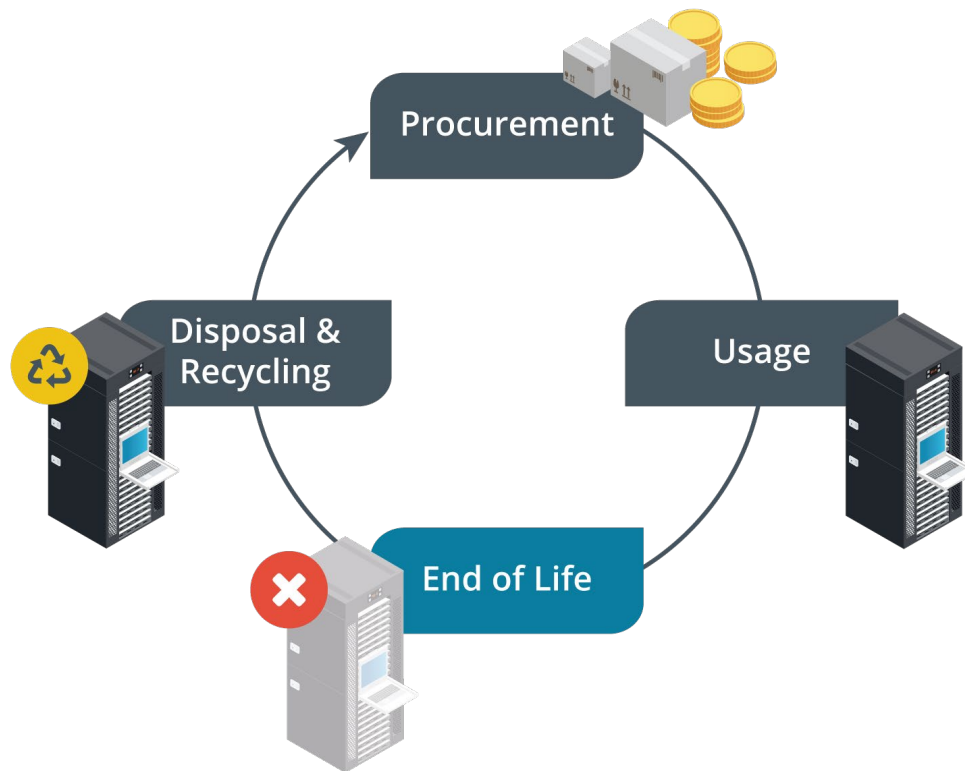
Server procurement. (Images © 123RF.com)

Usage



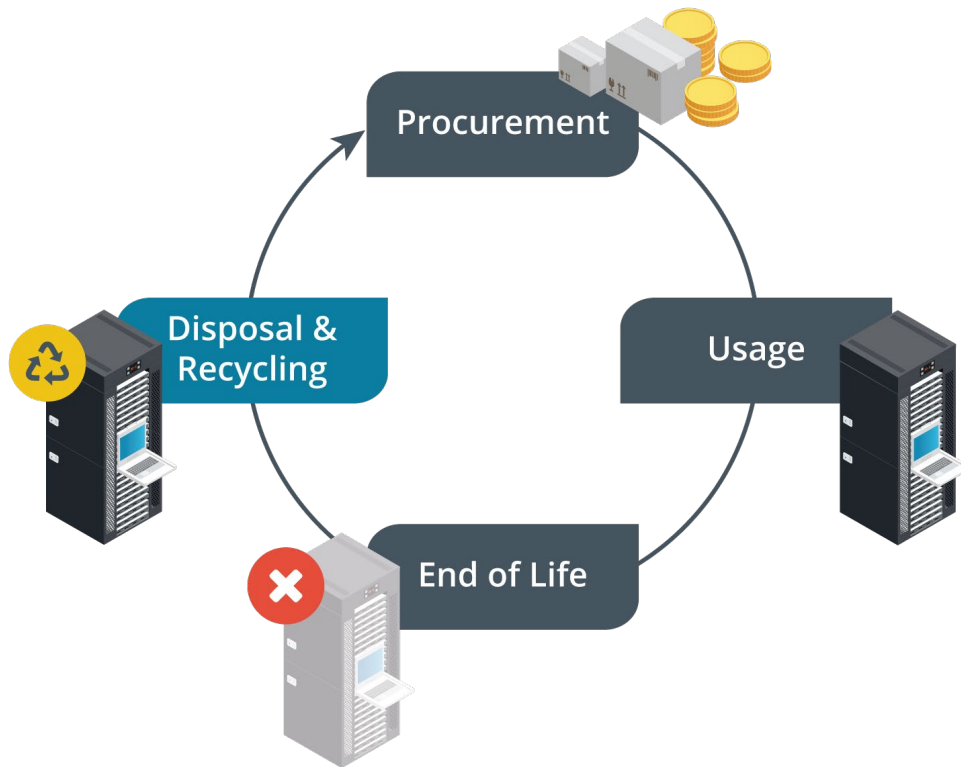
Server usage. (Images © 123RF.com)

End of Life



End of server life. (Images © 123RF.com)

Disposal/Recycling



Disposing of discarded server. (Images © 123RF.com)

Leasing Versus Owning and Service Plans

- Possible leasing benefits
 - Reduced Cap/Ex
 - Lower TCO
 - Simple budget--predictable payments
 - No hardware obsolescence
 - Possible ownership at end of lease
- Warranty and Service Plans

Inventory Management

- Importance of inventory management
- Risks of not tracking assets
- Benefits of tracking assets
- Asset management system features
- Tracking servers

Review Activity: Asset Management Concepts



Review Activity

Topic 4B

Manage Documentation

Exam Objectives Covered

- 2.3 Given a scenario, configure and maintain server functions and features
- 2.7 Explain the importance of asset management and documentation

Configurations and Baselines

- Standardize with templates:

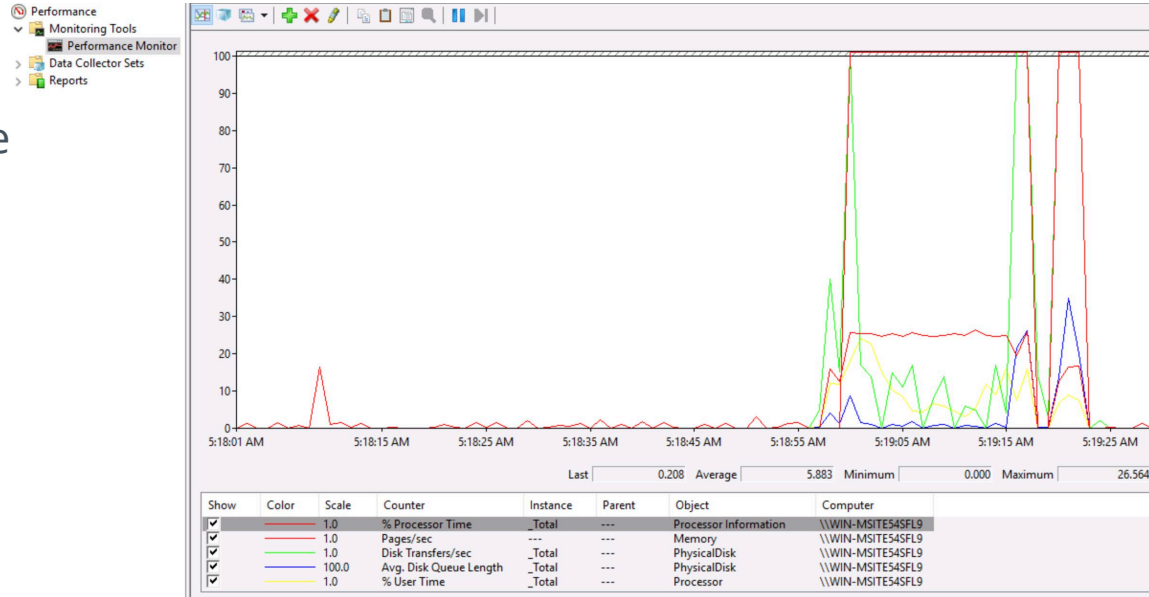
```
[root@CentOS7-A ~]# virt-install --name=LinuxSvr1 --vcpus=1 --memory=1024 --disk size=10 --cdrom=/media/cdrom/rhel8 --os-variant=rhel8
```

Use virt-install to define a virtual machine configuration.

- Standardized components
 - OS versions
 - Security software
 - Configuration of:
 - Local accounts
 - Storage and RAID arrays
 - BIOS passwords

Microsoft Performance Monitor

- Establish baselines
 - Processor--% User Time
 - Processor--% Processor Time
 - Memory—Pages/sec
 - Network Interface—Transfers/sec
 - Physical Disk—Disk Transfers/sec
 - Physical Disk—Average disk queue length

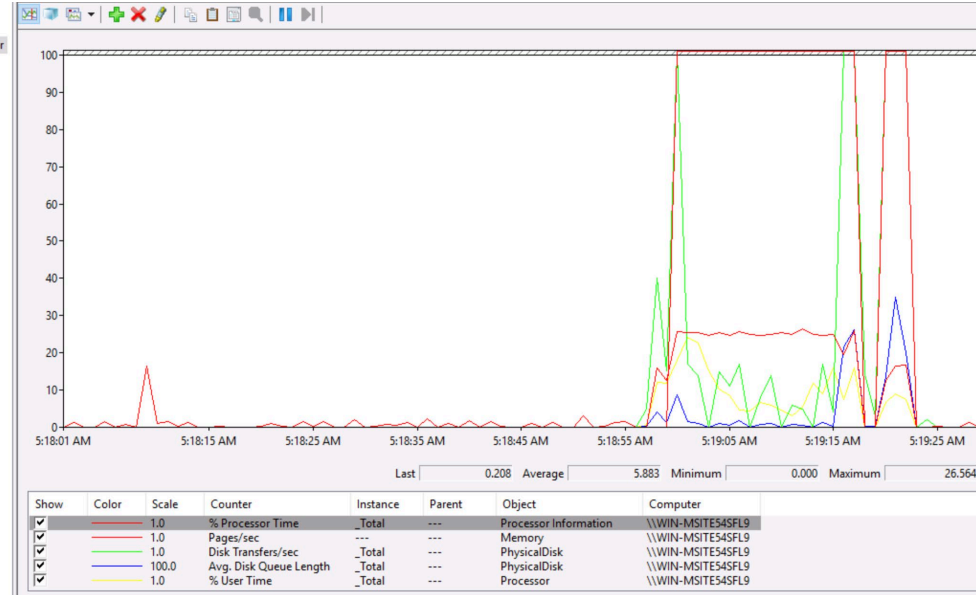
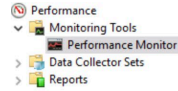


Performance Monitor with server baseline counters. (Screenshot courtesy of Microsoft.)

Microsoft Performance Monitor (slide 2)

- Real-time performance tools

- Windows
 - Task Manager
 - Resource Monitor
 - Reliability Monitor
- Linux
 - Sar
 - Top
 - Uptime



Performance Monitor with server baseline counters. (Screenshot courtesy of Microsoft.)

Design Documents

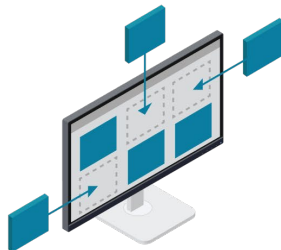
- Useful for:
 - Troubleshooting
 - Capacity planning

LAMP Server

Linux Operating System



Apache
(Web Server)



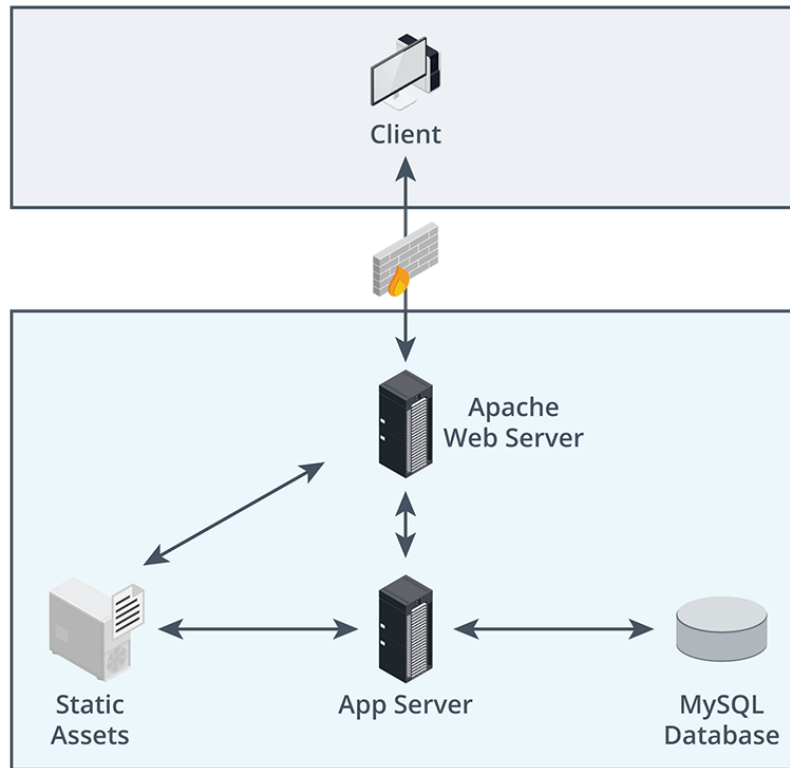
PHP
(Programming
Language)



MySQL
(Database)

LAMP server stack design document. (Images © 123RF.com)

Web Application



Architectural diagram for a web application. (Images © 123RF.com)

Standard Operating Procedure (SOP)

- Common elements:
 - Clear, descriptive title
 - Purpose statement
 - Author/owner with contact info
 - Scope
 - Detailed instructions
- Examples:
 - Server deployment/decommissioning
 - Building workstations
 - OS installation
 - User account lifecycle

Business Continuity Plan

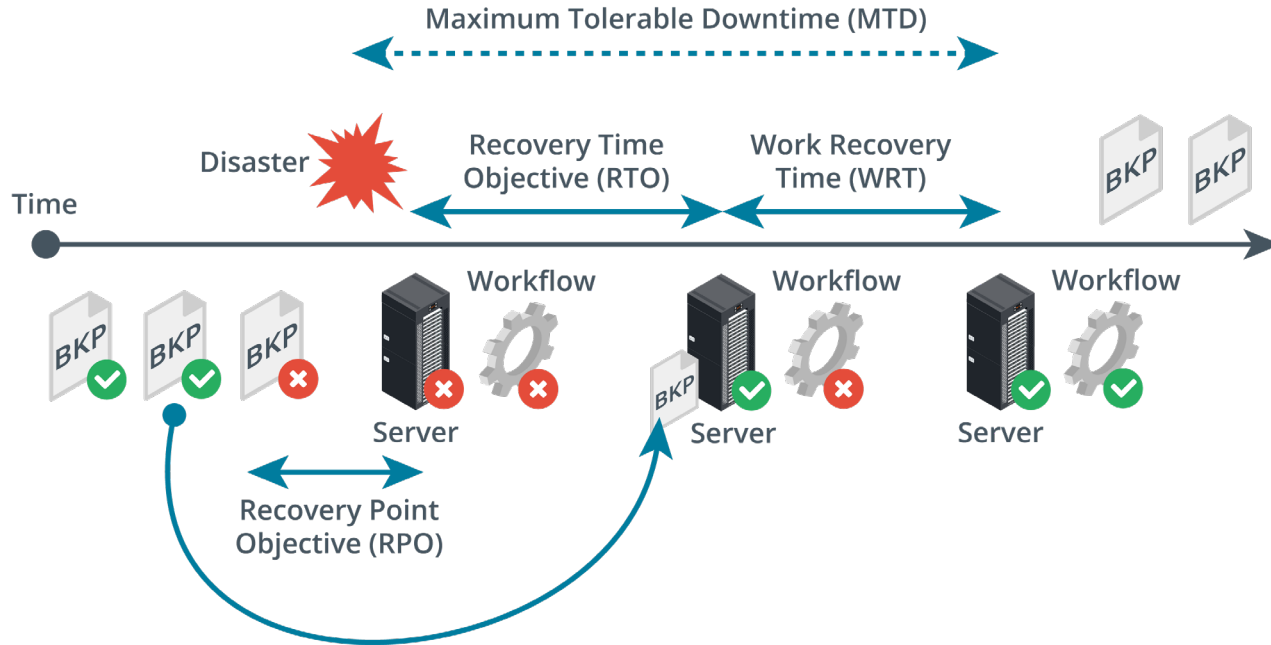
- BIA potential impacts:
 - Income
 - Reputation
 - Penalties
 - Expenses

Disaster Recovery Plan (DRP)

- Mean Time Between Failures (MTBF)
- Mean Time to Repair/Replace/Recovery (MTTR)

Disaster Recovery Plan (DRP) (slide 2)

- Recovery Point Objective (RPO)



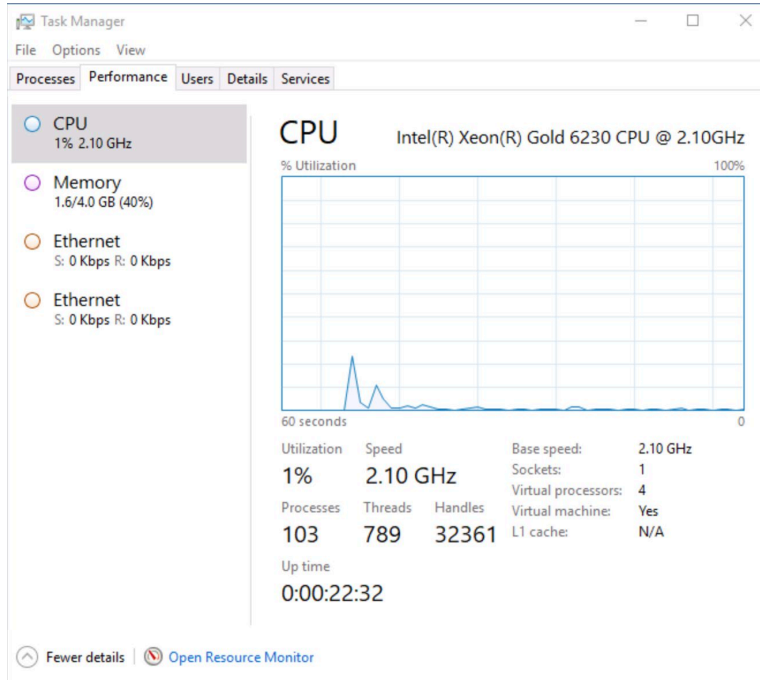
Disaster recovery – note RPO and RTO information. (Images © 123RF.com)

Disaster Recovery Plan (DRP) (slide 3)

- Recovery Time Objective (RTO)
- Service Level Agreement (SLA)

Uptime Requirements

- Uptime versus availability



Task Manager, Up time displayed at the bottom. (Screenshot courtesy of Microsoft.)

```
[root@localhost ~]# uptime
12:24:12 up 1:29, 1 user, load average: 0.00, 0.00, 0.00
[root@localhost ~]# █
```

The uptime command in Linux.

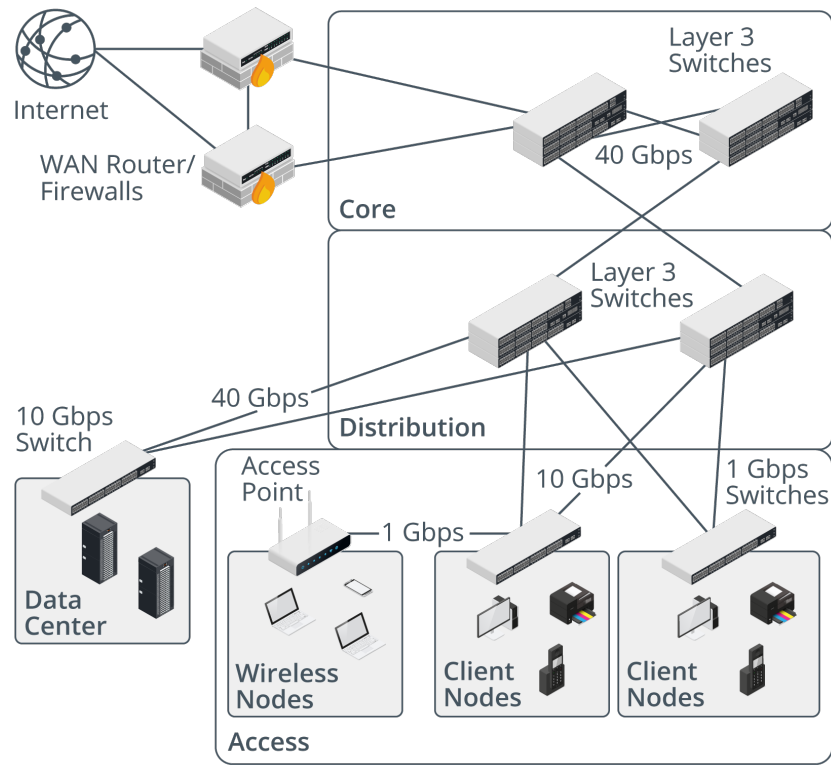
Troubleshooting Documents

- Service manuals
- Infrastructure diagrams
- Workflow diagrams
- Change management documents
- Document storage, access, and security

Troubleshooting Documents (slide 2)

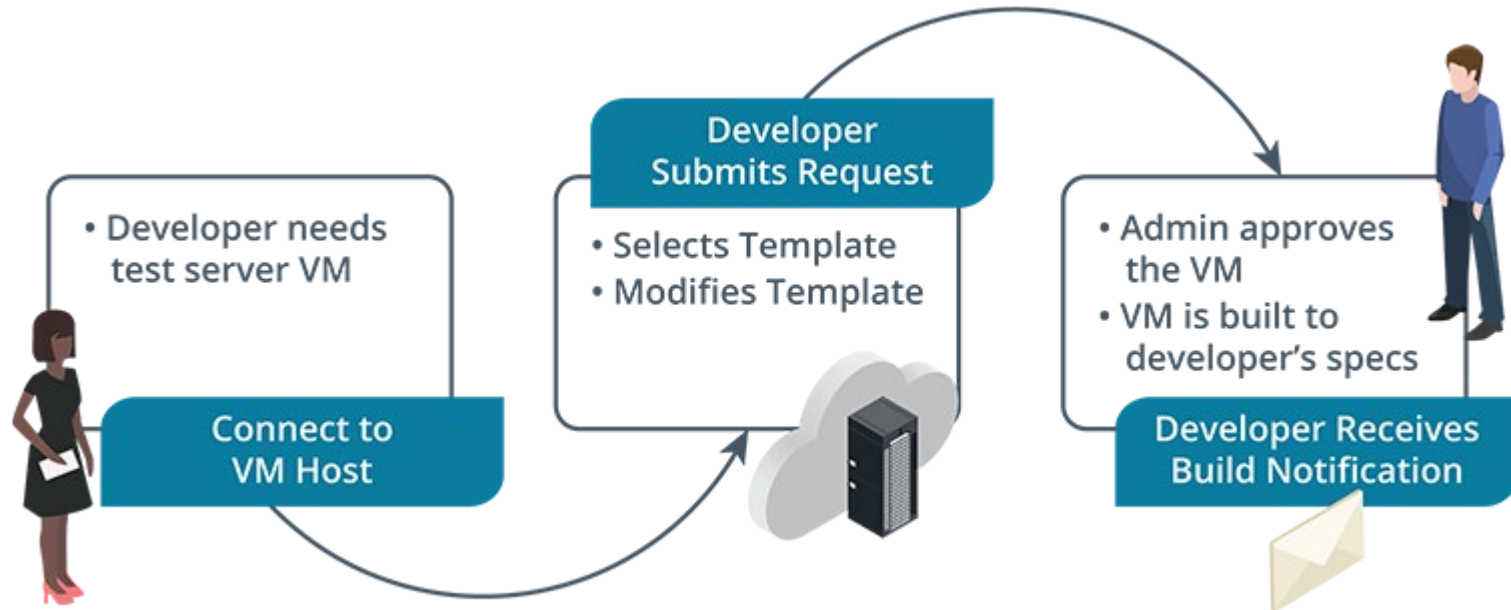
- Service manuals
 - Specs
 - Diagrams
 - Part numbers

Troubleshooting Documents (slide 3)



Network infrastructure diagram. (Images © 123RF.com)

Troubleshooting Documents (slide 4)



Server deployment workflow. (Images © 123RF.com)

Troubleshooting Documents (slide 5)

- Change management documents
 - Standardized
 - OS and application updates

Troubleshooting Documents (slide 6)

- Document storage, Access, and Security
 - Capture
 - Store
 - Locate
 - Update
 - Share

Guidelines for Managing Documentation

- Accurate, secure, and accessible
- Built-in tools
- Architectural documents
- BIA documents
- MTBF
- MTTR
- RPO
- RTO
- SLAs
- Infrastructure diagrams
- Workflow diagrams
- Change management

Review Activity: Documentation



Review Activity

Lesson 4

Summary

